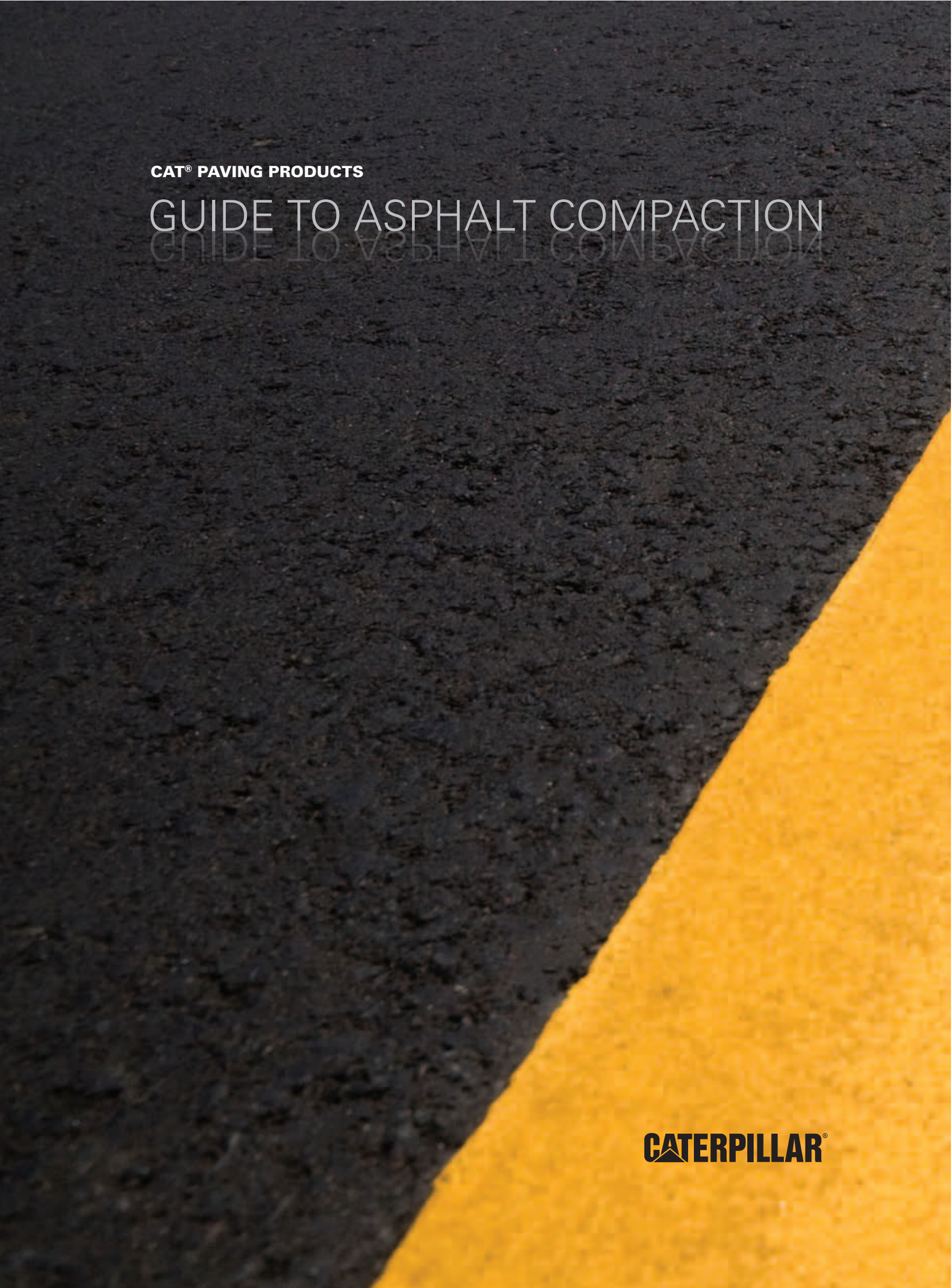
CAT® PAVING PRODUCTS

GUIDE TO ASPHALT COMPACTION



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GUIDE TO ASPHALT COMPACTION

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INTRODUCTION

The Cat Paving Products *Guide to Asphalt Compaction* is a practical, hands-on reference manual to be used by machine operators, quality control personnel and supervisors. It covers the basic principles of asphalt compaction and provides specific examples of how to use those principles most effectively.

In this guide, the word “asphalt” will be used to describe what may be called “bituminous material” or “asphalt concrete” in some parts of the world. Individual asphalt design formulas are discussed in Unit 3 and those will be referred to using commonly accepted terminology.

The design and manufacture of asphalt varies greatly around the world. Sources of aggregate are different. Likewise, asphalt cements used in the manufacture of asphalt have significant chemical variability. Finally, the types of paving equipment used to lay down asphalt are different in different locations. Therefore, with all this variability, it is impossible to develop specific and detailed asphalt compaction techniques that can be employed in all situations.

However, the principles of asphalt compaction remain the same for all applications. Equipment operators and quality control personnel must have a good understanding of these principles. They must know how to respond to the many variables.


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Unit 1
THE BASICS OF COMPACTION

Your crew can develop analytical skills that all but guarantee successful asphalt compaction. It starts with a commitment to fundamental best practices and proper planning.

[WHAT IS COMPACTION?]

Asphalt compaction is a mechanical process. Various forces are used to make the asphalt layer denser after the asphalt paver has laid it down. The purpose of compaction is to reduce the amount of air voids in the asphalt layer and to move the aggregate in the layer closer together. Strength is built into the asphalt layer by removing most of the air voids and developing stone-on-stone contact.

Usually compaction starts at the highest temperature possible. Once the asphalt layer has cooled below a given temperature, additional density is difficult or impossible to achieve. Therefore, only a short window of opportunity is normally available to create the required density. Planning and preparation are extremely important for the compaction of asphalt in order to complete the work in a timely manner.

For example, the density of an asphalt layer after it passes under a vibratory screed may be 85% of Theoretical Maximum Density. On another project using a different mix design and the same paving equipment, the density of the asphalt layer may be as low as 78% of Theoretical Maximum Density. Or, using paving equipment with tamping and vibrating screeds that deliver more energy to the asphalt layer, the screed laid density may be as high as 92% of Theoretical Maximum Density.

It is obvious that the compaction process will be different on each of these projects, even though the specified final density may be the same. Operators, quality control personnel and project supervisors must plan each project differently. It is likely that the type and number of compactors will be different. Rolling patterns also will be different.

In some places, public works officials have created method specifications for the compaction process. In those situations the compaction team must follow the specified procedure.

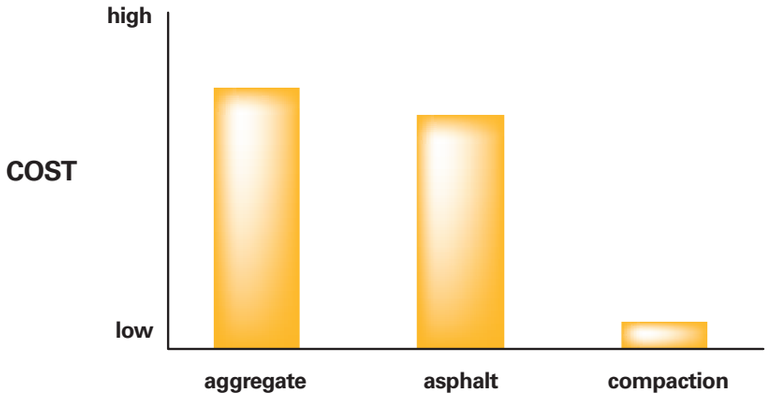
In other places, however, the public works department provides an end-result specification. In those situations, the compaction team is free to develop its own compaction method. For the most part in this guide, an assumption is made that the crew is working toward an end-result specification. A typical method specification is shown in Unit 4.





Asphalt compaction is started by the paver screed and finished by compactors working immediately behind the paver. Crew members often monitor the compaction process.

RELATIVE COST COMPARISON BETWEEN ASPHALT PAVEMENT COMPONENTS



The compaction process costs very little compared to the cost of aggregate and asphalt cement.

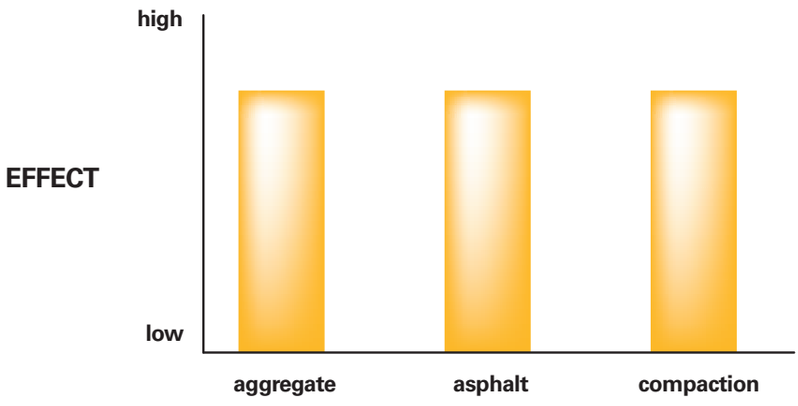
[THE VALUE OF COMPACTION]

Many years ago, asphalt compaction was considered a necessary evil and the process did not really add value to the pavement structure. In many instances, compactor operators were considered to be at the entry level of operating skills and typically were provided with very little training.

In recent years, the cost of quality aggregates has risen while their availability has declined. The price of asphalt cement has gone up even more. So, the cost relationship between materials and the compaction process places more emphasis on material production and laydown. Compaction actually costs very little per ton of bituminous material.

However, without the specified density, the bituminous material produced and laid down is virtually worthless. The compaction process must be viewed as equally important as production and laydown of material. Compactor operators need training to develop the necessary skills. Quality control personnel must be able to plan the compaction process and to solve problems when there is inadequate density or when the compaction process creates roughness in the surface layer.

RELATIVE COST COMPARISON BETWEEN EACH COMPONENT'S CONTRIBUTION TO EXTENDED PAVEMENT LIFE



Compaction has the same value as the material being produced.

[COMPACTION AIDS IN SUSTAINABILITY]

Properly compacted asphalt structures contribute to sustainability in several ways.

First, an asphalt structure is designed to support traffic volume and load over a given period of time. The design engineer calculates type of material, depths of layers and specified density in each layer to create the needed overall strength.

When the asphalt layers have consistently high density, the structure will usually provide the expected service life or even a service life that exceeds the plan.

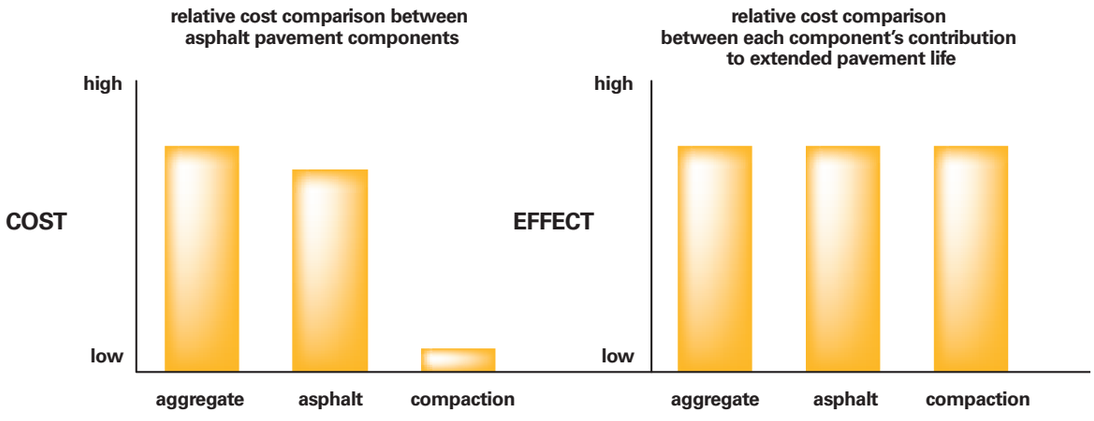
When pavement maintenance is delayed because the pavement remains in good condition, energy is saved (fewer emissions) and there is less traffic disruption (inconvenience and higher emissions) over the life of the structure. It is possible that one maintenance procedure may be eliminated over the life of the structure.

Second, highly-skilled compactor operators know how to create density in the asphalt layer with minimal influence on the smoothness of the surface layer. Plus, high density helps minimize surface defects such as rutting and cracking.

A smooth surface reduces rolling resistance and it takes less energy to propel a given vehicle over the surface at a given speed.

Even if fuel consumption improves by only 1% due to pavement smoothness, the global impact is enormous.

COMPONENT COSTS VS IMPACT ON PAVEMENT LIFE



A close comparison between cost and impact on pavement life.

Summary: *With so much riding on the creation of specified density in asphalt layers, it is important for operators to be trained, for quality control personnel to have problem-solving skills, and for technologically advanced equipment to be used.*

Compaction problems or issues are usually solvable when the crew plans well and applies fundamental best practices. The goal of this guide is to help all personnel involved in the compaction process to develop those analytical skills and a good working knowledge of best practices.

Unit 2 starts with an explanation of the forces that impact asphalt compaction.


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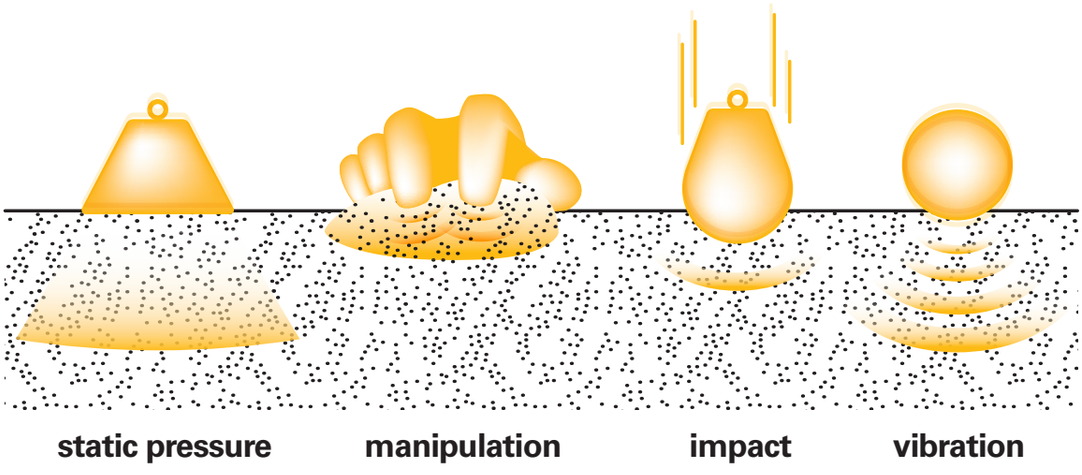
Unit 2
THE FORCES OF COMPACTION

Successful crews understand the relationship between the forces of compaction—and how a mat accepts those forces.

Four forces are used to expel air voids and create support in asphalt layers: static load, manipulation, impact and vibration. Machine operators and quality control personnel must understand how to use the four forces in order to create the required density in a productive manner while maintaining the smoothness of the asphalt layer.

Static load and manipulation usually involve lower forces and are the easiest to understand. Static load is created by a steel drum compactor operated in the static mode or by a pneumatic compactor.

Impact and vibration are dynamic forces and typically generate higher compaction forces. Vibratory steel drum compactors develop impact and vibration forces and typically receive most of the attention.



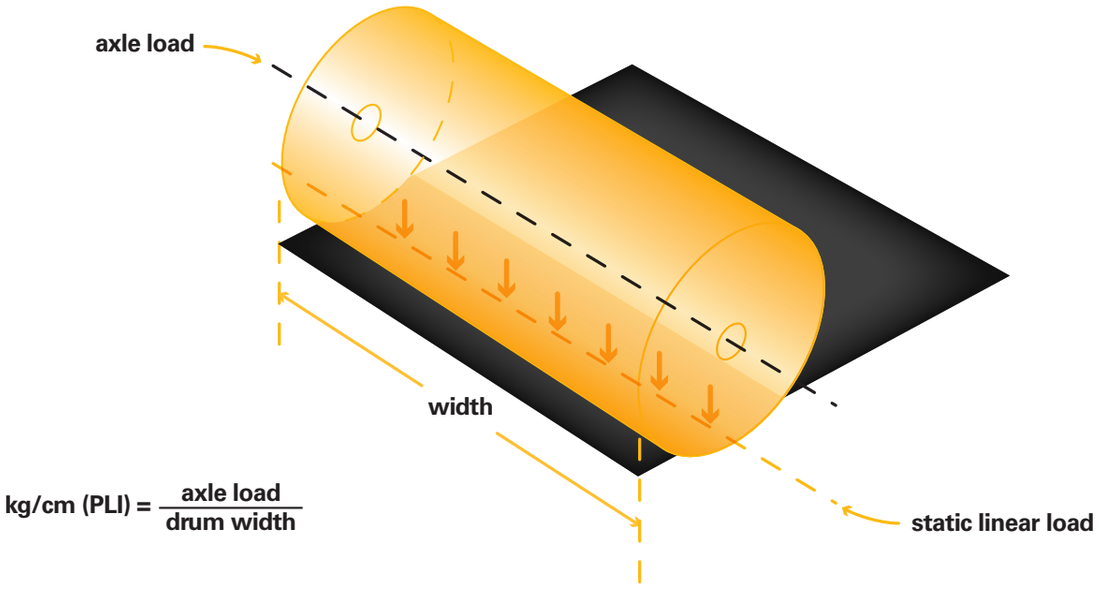
[STEEL DRUM STATIC PRESSURE]

Steel drum compactors operated in the non-vibratory mode apply static pressure on the asphalt mat. The amount of static pressure depends on the weight at the drum and the area of the drum that actually contacts the mat. A higher weight at the drum produces higher static pressure. Likewise a smaller contact area yields higher pressure. Static pressure is rated in bar or pounds per square inch (psi).

A simpler way to look at static force is to divide weight at the drum by width of the drum. This result is expressed as kilograms per centimeter or pounds per inch. It is important to remember that the highest machine weight does not always produce the highest static load.

The chart below shows three Cat Tandem Vibratory Rollers. The heaviest machine is the CB64, a unit with 213 cm (84") wide drums. The next unit, the CB54XW, weighs less and has drums that are 200 cm (79") wide. The lightest machine is the CB54 with 170 cm (67") drums.

	CB64	CB54XW	CB54
Weight @ drum	6490 kg	5949 kg	5402 kg
Drum width	213 cm	200 cm	170 cm
Static linear load	31 kg/cm	30 kg/cm	32 kg/cm



$$\text{kg/cm (PLI)} = \frac{\text{axle load}}{\text{drum width}}$$

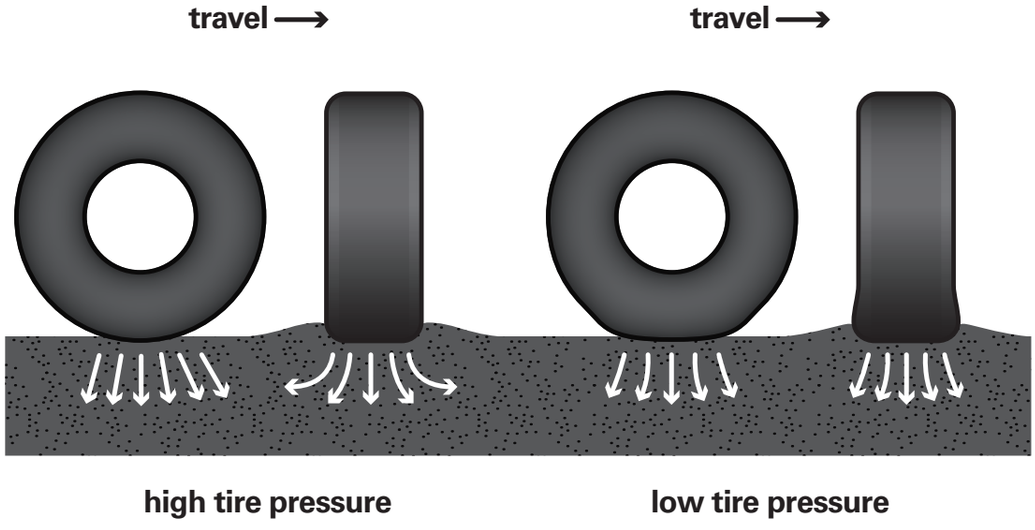
What is interesting to note is that the lightest unit, the CB54, has the highest static linear load. This is often the case because the drums are narrower.

Therefore, if you are working on a project and you need a compactor to deliver a relatively high static force, then you will probably want to use the narrowest drum model available to you that can still match the production requirement.

So, to summarize, classify all your steel drum compactors by static linear load. This knowledge will help you select the right equipment for static applications.

User Tip: *The finish compaction phase is usually accomplished by a steel drum compactor set in the static mode. A compactor with relatively narrow drums and which has a higher linear load will clean up drum stop marks better than a compactor with wider drums and less linear load. The compactor with narrower drums and higher linear load may even achieve a little more density in the finish phase.*

VARIABLE TIRE PRESSURE EFFECT ON COMPACTION FORCE



[PNEUMATIC COMPACTOR STATIC PRESSURE]

The other type of compactor which exerts static force is a pneumatic, or rubber-tire roller. The amount of ground pressure depends on the weight at each tire and the area of the tire in contact with the mat.

You can change the weight at each tire by changing the amount of ballast in the compactor. Adding weight increases the load per tire and static force will penetrate deeper into the mat.

Most pneumatic compactors have ballast tanks that are filled with water, or wet sand, or other material. Some pneumatic compactors come with optional, removable steel weights. Once the pneumatic compactor has been delivered to the jobsite, its ballasted weight is seldom changed. At the jobsite, the common way to change the amount of static force is by adjusting tire pressure.

When the tire pressure is decreased, the tire flattens more and the contact area is greater. Therefore, less static pressure is applied to the mat.

CW34 BALLASTED TO 2000 KG AND 3000 KG

	Ground Contact Pressure	
Tire Pressure	**@ 2000 kg**	**@ 3000 kg**
300 kpa (44 psi)	260 kpa (38 psi)	397 kpa (58 psi)
500 kpa (73 psi)	357 kpa (52 psi)	386 kpa (56 psi)
700 kpa (102 psi)	498 kpa (72 psi)	457 kpa (66 psi)
900 kpa (131 psi)	764 kpa (111 psi)	573 kpa (83 psi)



The air-on-the-run option makes it easier to adjust tire inflation pressure.

When the tire pressure is increased, the tire stands up straighter and the mat contact area is less. A smaller contact area results in higher static force applied to the mat.

There are two things to be aware of when increasing tire pressure. First, the higher static force will leave deeper cut marks in the surface of the mat. These deep cut marks may be difficult to clean up during the finish compaction phase.

Second, never exceed the manufacturer's recommended maximum tire inflation pressure. Over-inflating rubber tires can cause premature tire failure.

User Tip: *When checking and adjusting tire pressures, be sure to inflate each tire to the same pressure. If tire pressures are variable, density across the mat will also be variable. In addition, you may notice that hot asphalt sticks more readily to the under-inflated tire or tires. Tire maintenance and inspection are critical on pneumatic compactors.*

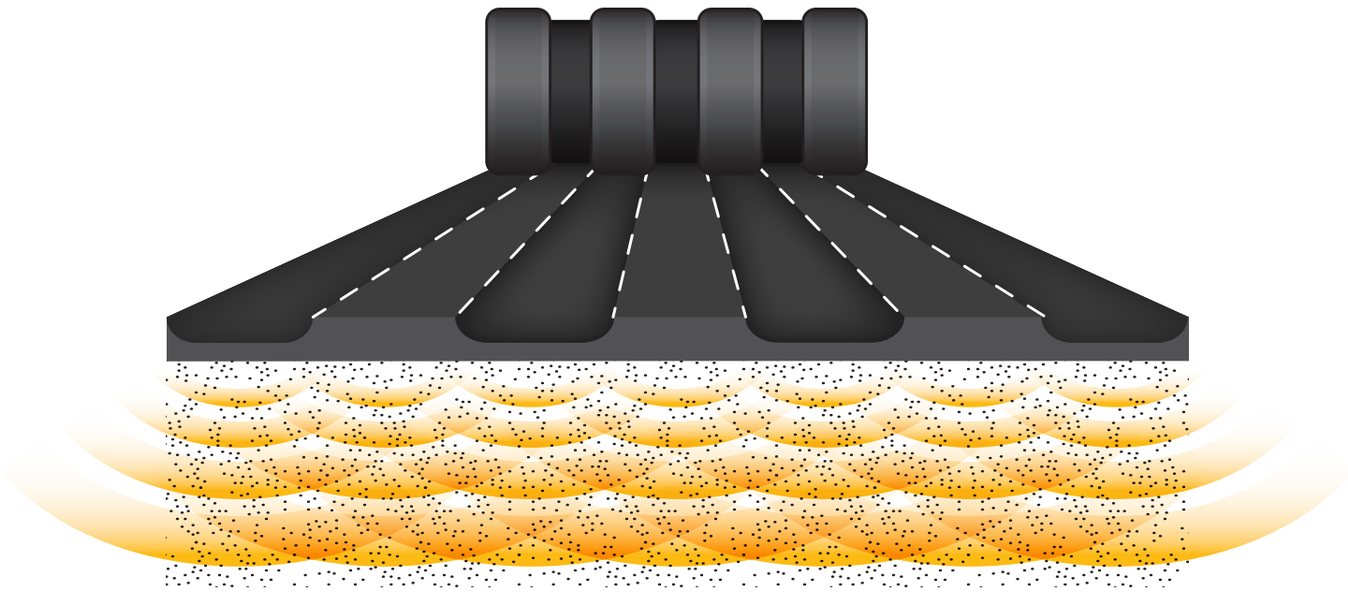
User Tip: *Static force exerted by a steel drum compactor or pneumatic compactor is affected by the working speed of the machine. The faster the working speed, the lower the density. Therefore, if you need higher density behind a static compactor, your first change should be to a slower working speed. You can add more passes, too, but a slower working speed should be the first variable that you consider.*

[MANIPULATION]

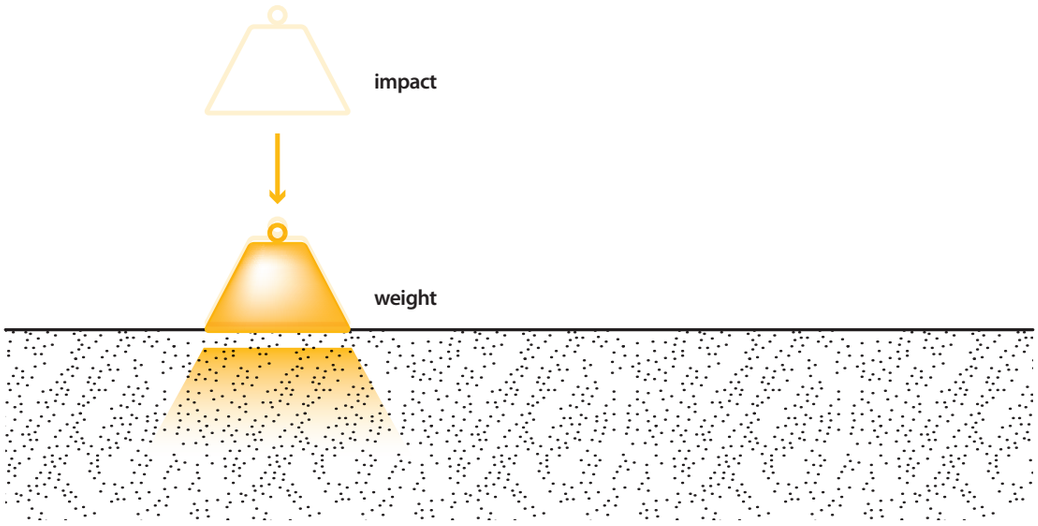
Manipulation, which is also a static force, occurs when the forces exerted into the mat are not entirely vertical. Instead, the lines of force are sent in many directions. The benefit of manipulation is that this force changes the surface texture by making it tighter. Manipulation is associated with pneumatic compactors and oscillatory compactors.

The staggered, overlapping tires on the axles of pneumatic compactors manipulate the mat under and between the tires in a confined manner. The lines of force are not only vertical but also move sideways. The vertical forces push down the large aggregate to increase density while the side-to-side forces create a tighter surface finish that helps prevent moisture penetration.

Some compactors have oscillatory drums. Oscillatory drums create tangential, or back-and-forth, lines of force that work primarily on the surface of the mat. Oscillatory force produces the same benefit of manipulation. It tightens and seals the surface of the mat.



Overlapping tires develop overlapping areas of contact pressure, creating manipulation forces.



[IMPACT]

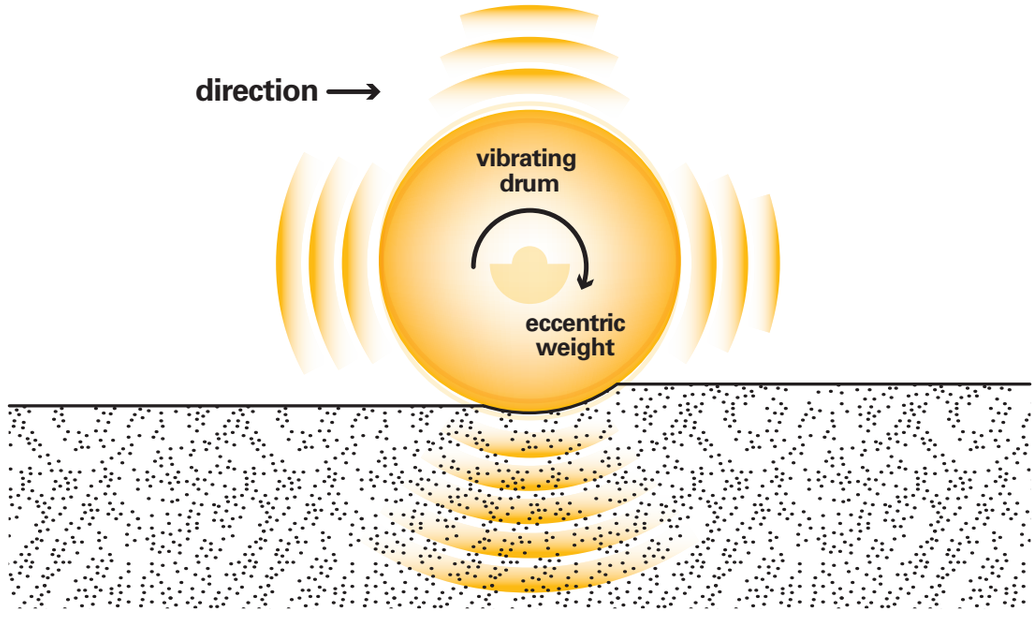
Impact, the next compaction force, is dynamic and creates more force on the mat than an equivalent static load. You have already learned that the weight at the drum divided by the width of the drum produces a static linear load.

On vibratory steel drum compactors, the drum actually moves into the mat. The static force of the drum is increased by drum movement or drum impact. The impact generates more energy. The impact energy is strongest at the surface of the mat and diminishes as it penetrates deeper into the mat.

Impact forces create density in the mat faster than static forces. Increased production is the benefit of using vibratory steel drum compactors.

The risk of using impact force is that too much energy may damage the aggregates in the mat. It is possible to over-compact the mat by using too much impact force; mat density can actually decrease when too much force is applied. What is needed for effective compaction is a balance between impact forces and other machine characteristics like weight, working speed and vibration frequency.

VIBRATION



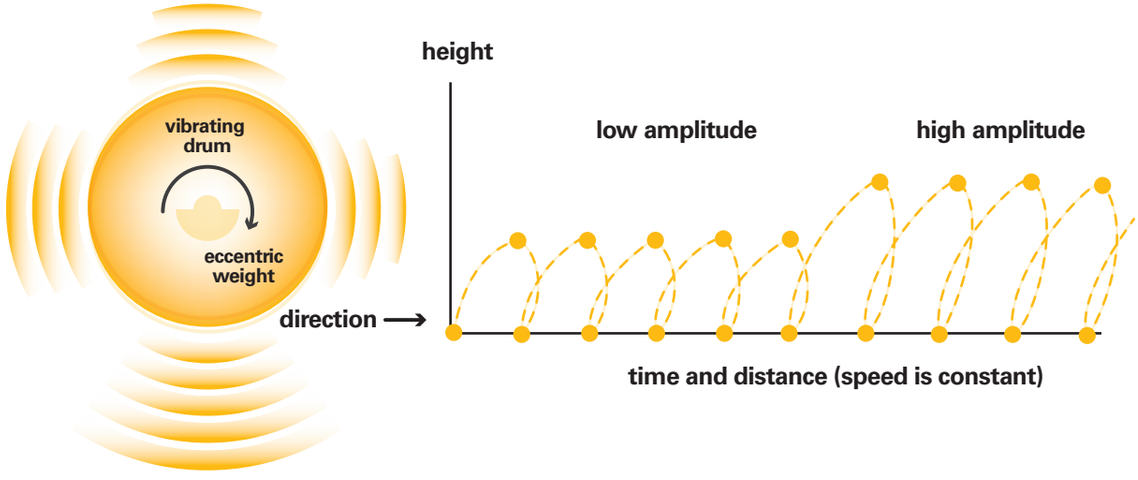
[VIBRATION]

The force of vibration is the most complex of the four compaction forces. Vibratory forces increase the energy developed by weight and impact.

A vibratory shaft is inside the steel drum. On the center of the vibratory shaft, there is an eccentric weight. When the vibratory system is activated, the vibratory shaft begins to spin rapidly. The rotation of the eccentric weight shaft causes the drum to move, or vibrate, in all directions. Vibration causes a series of pressure waves to be released into the mat. The vibratory pressure waves cause the aggregates in the mat to move. Aggregate movement helps to reorient the larger aggregates so the impact force can more easily reduce the air voids between the aggregates, thereby locking them into contact position.

User Tip: *In general, select the highest amplitude that will be accepted by the mat without causing the drums to bounce or creating impact marks. Remember, amplitude selection has the greatest impact on the creation of density and, therefore, the production rate of the compactor.*

AMPLITUDE



[IMPACT EQUALS AMPLITUDE]

On a steel drum vibratory compactor, you have learned that the drums move up and down rapidly to create impact and vibration. The impact force caused by the drum moving into the mat is rated by the term known as amplitude.

Amplitude is the distance that the drum moves into the mat. Amplitude is the most significant factor when discussing compaction effectiveness.

Most vibratory compactors offer a variety of amplitude settings. When the amplitude is changed by the operator, the configuration of the eccentric weight inside the drum is changed. When the eccentric weight is the most off-center, amplitude is the highest and impact force is increased. When the eccentric weight is more balanced, the amplitude is decreased and the impact force is smaller.

All personnel involved in the compaction process should know the amplitude capabilities of each compactor on the job. They should be able to develop a checklist to help them select the correct amplitude, if required.

In general there are three amplitude ranges: low, medium and high.

Low Amplitude Range	0.2 mm to 0.5 mm (0.01" - 0.02")
Medium Amplitude Range	0.5 mm to 0.8 mm (0.02" - 0.03")
High Amplitude Range	Above 0.8 mm (0.03")